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1 *drop missing vae cd4lt*
2 drop if missing(cd4lt)
3 .
4 *check missing*
5 count if missing(cd4bl)
6 .
7 *drop extreme value cd4bl*
8 drop if cd4bl == 5022 | cd4bl == 6740
9 .
10 *ageartg label*
11 label define ageartg_lbl 0 "15-24" 1 "25-49" 2 ">=50",modify
12 label values ageartg ageartg_lbl
13 .
14 *generate cd4bl group & label value*
15 generate cd4blg = .
16 replace cd4blg = 0 if cd4bl < 200
17 replace cd4blg = 1 if cd4bl >= 200 & cd4bl < 350
18 replace cd4blg = 2 if cd4bl >= 350
19 replace cd4blg = 3 if missing(cd4bl)
20 .
21 label define cd4blg_lbl 0 "<200" 1 "200-<350" 2 ">=350" 3 "Unknown",modify
22 label values cd4blg cd4blg_lbl
23 .
24 *check date var*
25 describe firstd diagd farvdate lastd
26 .
27 *generate Duration of art*
28 generate du_art = lastd - farvdate
29 .
30 *generate Duration of art group*
31 generate du_art_g = du_art
32 replace du_art_g = 0 if du_art <= 182.625
33 replace du_art_g = 1 if du_art > 182.625 & du_art <= 365.25
34 replace du_art_g = 2 if du_art > 365.25
35 .
36 label define du_artg_lbl 0 "Within 6 months" 1 "> 6 to 12 months" 2 "> 12
months",modify
37 label values du_art_g du_artg_lbl
38 .
39 *generate diagd_n*
40 generate diagd_n = diagd
41 replace diagd_n = firstd if missing(diagd)
42 format diagd_n %td
43 .
44 *generate Rapid of art*
45 generate rapid_art = farvdate - diagd_n
46 .
47 *generate Rapid of art group for Characteristics*
48 generate rapid_artg_c = rapid_art
49 replace rapid_artg_c = 0 if rapid_art <= 30.4375
50 replace rapid_artg_c = 1 if rapid_art > 30.4375
51 .
52 label define rapid_artg_c_lbl 0 "Rapid" 1 "Not Rapid" ,modify
53 label values rapid_artg_c rapid_artg_c_lbl
54 .
55 *generate Rapid of art group for factor*
56 generate rapid_artg = rapid_art
57 replace rapid_artg = 0 if rapid_art <= 30.4375
58 replace rapid_artg = 1 if rapid_art > 30.4375 & rapid_art <= 91.3125
59 replace rapid_artg = 2 if rapid_art > 91.3125
60 .
61 label define rapid_artg_lbl 0 "Within 1 month" 1 "> 1 to 3 months" 2 "> 3
months",modify
62 label values rapid_artg rapid_artg_lbl
63 .
64 *drop rapid_art negative date*
65 drop if rapid_art < 0
66 .
67 *generate HIV Stage*
68 gen hivstage = infect_phase
69 replace hivstage = 1 if infect_phase ==4
70 replace hivstage = 1 if infect_phase ==5
71 replace hivstage = 2 if infect_phase ==6
72 replace hivstage = 3 if infect_phase ==7
73 replace hivstage = 4 if hivstage == .

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74 replace hivstage = 3 if firstoi=="Y"
75 replace hivstage = 1 if firstoi=="N" & hivstage == 4
76
77 label define hivstage 1 "Asymptomatic HIV" 2 "Symptomatic HIV" 3 "AIDS"
78 label values hivstage hivstage
79 .
80 *generate fartg*
81 gen fartg = 1 if index(fregimen,"NNRTI") > 0 & index(fregimen,"PIs") == 0
82 replace fartg = 2 if index(fregimen,"NNRTI") == 0 & index(fregimen,"PIs") >0
83 replace fartg = 3 if index(fregimen,"DTG")
84 replace fartg = 4 if index(fregimen,"NNRTI") == 0 & index(fregimen,"PIs") ==
0 & index(fregimen,"NRTI") > 0
85
86 replace fartg = 5 if index(fregimen,"NNRTI") > 0 & index(fregimen,"PIs") > 0
87 replace fartg = 6 if fartg == .
88 replace fartg = 7 if fdrug == ""
89
90
91 replace fartg = 2 if fartg == 4 & lregimen=="PIs+NRTI"
92 replace fartg = 2 if fartg == 4 & lregimen=="mono-PIs"
93 replace fartg = 2 if fartg == 5 & lregimen=="PIs+NRTI"
94 replace fartg = 1 if fartg == 5 & lregimen=="NNRTI+NRTI"
95 replace fartg = 2 if fartg == 5 & lregimen=="mono-PIs"
96 replace fartg = 1 if fartg == 5 & lregimen=="mono-NNRTI"
97
98
99 label define fartg 1 "NNRTI" 2 "PI" 3 "DTG-based" 4 "only NRTI" 5 "NNRTI+PI" 6
"Third-line" 7 "unknown",replace
100 label values fartg fartg
101 .
102
103 *คำสั่ง list*
104 list fartg fregimen if fregimen == "INSTIs"
105 .
106 *เช็ค label*
107 tab fartg, nolabel
108 .
109 *generate fartg_n*
110 generate fartg_n = .
111 replace fartg_n = 1 if fartg == 1
112 replace fartg_n = 2 if fartg == 2
113 replace fartg_n = 3 if fartg == 3
114 replace fartg_n = 4 if inlist(fartg, 4, 5, 6)
115
116 label define fartgn_lbl 1 "NNRTI" 2 "PI" 3 "DTG-based" 4 "Others",modify
117 label values fartg_n fartgn_lbl
118 .
119 *ใส่ label ในตัวแปร died*
120 label define died_lbl 0 "No" 1 "Yes"
121 label values died died_lbl
122 .
123 *ใส่ label ในตัวแปร sex*
124 label define sex_lbl 1 "Male" 2 "Female"
125 label values sex sex_lbl
126 .
127 *generate firstoi_g*
128 generate firstoi_g = .
129 replace firstoi_g = 1 if firstoi == "Y"
130 replace firstoi_g = 0 if firstoi == "N"
131 label define firstoi_g_lbl 0 "No" 1 "Yes"
132 label values firstoi_g firstoi_g_lbl
133 .
134 *generate vly1 group*
135 generate vly1_g = .
136 replace vly1_g = 0 if vly1 <= 50
137 replace vly1_g = 1 if vly1 >50
138 replace vly1_g = 2 if missing(vly1)
139 .
140 label define vly1_g_lbl 0 "<=50" 1 ">50" 2 "Unknown",modify
141 label values vly1_g vly1_g_lbl
142 .
143 *จัดกลุ่ม province*
144 generate area_ng = .
145 replace area_ng = 0 if inlist(province, "Chiang Mai","Chiang Rai","Lamphun",
"Lampang", "Mae Hong Son","Phrae","Phayao","Nan")
146 replace area_ng = 1 if inlist(province, "Tak","Sukhothai","Phitsanulok",
"Phichit", "Kamphaeng Phet","Nakhon Sawan","Uthai Thani","Phetchabun",
"Uttaradit")

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147 label define area_ng_lbl 0 "Upper_North" 1 "Lower_North",modify
148 label values area_ng area_ng_lbl
149 .
150 *แทนค่าสูญหาย*
151 replace area_ng = 1 if missing(area_ng)
152 .
153 *cd4lt แบ่งกลุ่ม*
154 generate cd4ltg = .
155 replace cd4ltg = 1 if cd4lt <= 500
156 replace cd4ltg = 2 if cd4lt > 500
157 .
158 label define cd4ltg_lbl 0 "<=500" 1 ">500",modify
159 label values cd4ltg cd4ltg_lbl
160 .
161 gen HIV_diag = .
162
163 replace HIV_diag = 1 if inrange(year_diagd_n,2008,2013)
164 replace HIV_diag = 2 if inrange(year_diagd_n,2014,2016)
165 replace HIV_diag = 3 if inrange(year_diagd_n,2017,2019)
166 replace HIV_diag = 4 if inrange(year_diagd_n,2020,2022)
167
168 label define HIV_diag_lbl 1 "2008-2013" 2 "2014-2016" 3 "2017-2019" 4
    "2020-2022",modify
169 label values HIV_diag HIV_diag_lbl
170 .
171 gen yearart_n = .
172 .
173 replace yearart_n = 0 if yearart == 2021
174 replace yearart_n = 1 if yearart == 2022
175 .
176 label define yearart_n_lbl 0 "2021" 1 "2022",modify
177 label values yearart_n yearart_n_lbl
178 .
179 category data
180 foreach var in sex ageartg cd4blg vly1_g du_art_g rapid_artg hivstage fartg_n
    firstoi_g area_ng HIV_diag yearart_n {
181     regress cd4lt i.`var',vce(robust) baselevel
182 }
183 .
184 gen vly1g_g = .
185 replace vly1g_g = 0 if vly1_log < 3
186 replace vly1g_g = 1 if vly1_log >= 3 & vly1_log < 4
187 replace vly1g_g = 2 if vly1_log >= 4 & vly1_log < 5
188 replace vly1g_g = 3 if vly1_log >= 5
189 replace vly1g_g = 4 if missing(vly1_log)
190
191 * 4. กำหนด Label ให้ตัวเลขแสดงผลเป็นข้อความ
192 label define vlyyy_label 0 "<3" 1 "3-3.99" 2 "4-4.99" 3 ">5" 4 "Unknown"
193 label values vly1g_g vlyyy_label
194 .
195 *แบ่งกลุ่มประกัน
196 generate maininscl_g = .
197 replace maininscl_g = 0 if maininscl == "UCS"
198 replace maininscl_g = 1 if maininscl == "SSS"
199 replace maininscl_g = 3 if maininscl == ""
200 replace maininscl_g = 2 if maininscl != "UCS" & maininscl != "SSS" & maininscl
    != ""
201 .
202 label define maininscl_g_lbl 0 "UCS" 1 "SSS" 2 "Others" 3 "Unknown",modify
203 label values maininscl_g maininscl_g_lbl
204 .
205 generate du_art_m = du_art/30.4375
206 generate rapid_art_m = rapid_art/30.4375
207 .
208 stcox i.sex ib3.fartg_n ib1.firstoi_g i.yearart_n i.rapid_artg ib1.minc_group,
    tvc(i.ageartg i.cd4blg ib2.hivstage i.du_art_g i.area_ng) baselevel

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